Junke Zhao

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LinkedIn | Website

Research Interests

I study how AI can reshape the relationship between people and the spaces and objects they live with, along two complementary directions:

- **Spaces that Understand:** environments and everyday objects that use AI to sense and reason about people's needs, then proactively respond to and serve them, such as transformable furniture agents that learn daily routines and decide when to act on their own, ask first, or wait.
  Keywords: Agentic Objects, Human-Agent Interaction, Shape-Changing Interfaces, Ubiquitous Computing
- **People who Shape:** AI-assisted design and fabrication tools through which people, from non-experts to designers, customize and improve their own environments, such as matching everyday needs to transformable furniture, human-AI co-creation in fabrication, and design editors for e-textiles.
  Keywords: Human-AI Co-Creation, Personal Fabrication, Computational Design, Creativity Support Tools

Education

The University of Texas at Dallas — Richardson, TX
Doctor of Philosophy, Computer Science — August 2026 - Present
Advisor: Liang He

Carnegie Mellon University — Pittsburgh, PA
Master of Science, Computational Design — August 2023 - May 2025
Thesis: CoFAB: An Integrated Human-AI Collaboration Prototyping and Fabrication System for Designers in Early Design Stages (Advisor: Joshua Bard)
Honors: Merit Scholarship

Southeast University — Nanjing, China
Bachelor of Architecture, Architecture — August 2018 - June 2023

ETH Zurich — Hybrid
Joint Program, Architecture — September 2019 - June 2020

Publications

- **Junke Zhao**, Te-Yen Wu, Liang He. *Beyond Furniture: A Taxonomy and Exploration of Transformable Furniture for Domestic Use*. Submitted to CHI 2027 (under review).
- **Junke Zhao**. *CoFAB: An Integrated Human-AI Collaboration Prototyping and Fabrication System for Designers in Early Design Stages*. Master's Thesis, Carnegie Mellon University, 2025.

RESEARCH EXPERIENCE

Department of Computer Science, The University of Texas at Dallas June 2025 - Present

Graduate Research Assistant | Advisor: Liang He, Te-Yen Wu (Florida State University) Richardson, TX

Beyond Furniture: A Taxonomy and Exploration of Transformable Furniture **(Submitted to CHI 2027)**

- Led a PRISMA-guided systematic review of 2,631 records from ACM DL, IEEE Xplore, Scopus, and Google Scholar, yielding 123 publications and 261 furniture items; built a taxonomy linking transformable mechanisms (behaviors, triggers, structures, materials), user profiles, and domestic uses
- Designed a hybrid human-LLM coding pipeline: two researchers coded a 14-paper gold set (Cohen's $\kappa = 0.77$), and a retrieval-augmented LLM pipeline coded the remaining corpus ($\kappa = 0.85$, 93.6% agreement with manual coding)
- Developed and deployed an AI-assisted exploration platform (React, Three.js, Flask, LangChain, Docker) that connects everyday requests to documented designs through LLM-generated explanations, animated 3D previews, and room placement
- Conducted a formative survey ($N = 48$) and an exploratory user study ($N = 12$) on how people discover and assess transformable furniture for their homes

Agentic Furniture: Transformable Furniture as an Autonomous Co-Inhabitant **(Ongoing)**

- Investigating when transformable furniture should act on its own, ask first, or wait, framing furniture as an agent that senses, remembers, reasons, and initiates its own physical transformations
- Built an LLM-driven simulation testbed (TypeScript, Three.js, OpenAI API) of a 54 m^2 studio apartment with desk and chair agents that combine sensing with safety gates, memory with reflection and routine learning, a confidence-gated act/ask/hold policy, and multi-agent arbitration
- Ran 180 simulated days (about 35,000 model calls) to compare autonomy policies; planning a formative study and a desktop-scale physical prototype

Design Editor for an Electronic Yarn Toolkit **(Ongoing, targeting UIST)**

- Leading software development of a Rhino 8 plugin (C#, RhinoCommon, Eto.Forms) that lets garment designers place electronic yarns on 2D pattern panels imported from CLO 3D and preview them on the draped 3D garment
- Implemented pattern import, barycentric 2D-to-3D mapping, a yarn library with design-rule checking, seam-crossing yarns, and reusable designs that re-apply to modified garments; developing strain visualization

School of Architecture, Carnegie Mellon University August 2024 - January 2025

Research Assistant | Design Fabrication Lab | Advisor: Joshua Bard Pittsburgh, PA

Robotic Shaping and Plastering of Hardset Building Materials

- Designed specialized end effectors for robotic plastering—variable-angle scraper fixture and RealSense camera detection system; developed corresponding automatic path generators in Grasshopper implementing linear paths, angle rotation, thickness-uniform cross paths, and automatic tool cleaning routines
- Executed manufacturing experiments with an ABB IRB 6700 robotic arm and optimized system performance; conducted systematic testing using varied scraper orientations and path strategies to enhance single-execution quality; managed experimental data collection and performance analysis
- Implemented a vision-based defect classification system using Arduino and Linux-connected RealSense cameras; automated surface defect identification and remedial path generation for robotic plastering operations

School of Architecture, Southeast University September 2020 - July 2021

Research Assistant | Institute of Building Technology and Science Nanjing, China

- Conducted comprehensive post-occupancy evaluation (POE) of climate-adaptive design strategies for an office building in Changzhou, China (hot summer/cold winter region); validated effectiveness of transitional space design methods for indoor microclimate regulation
- Implemented systematic environmental monitoring using positioned sensors over one year; collected quantitative data on temperature, humidity, airflow, solar radiation, and other indoor environmental parameters across multiple seasons; conducted satisfaction surveys capturing occupant comfort data throughout the year
- Developed detailed thermal performance, daylighting, and natural ventilation simulation models in Rhino and ClimateStudio; compared predicted performance against measured data to validate design methodologies; created comprehensive visualizations characterizing environmental conditions across different building zones and seasons

Skills

- **Programming Languages:** Python, C#, TypeScript/JavaScript, Grasshopper, Arduino, ROS, Processing
- **Design and Creative:** Photoshop, Illustrator, InDesign, Figma, Adobe Premiere
- **Modeling and Visualization:** Rhinoceros 3D, Grasshopper, SolidWorks, AutoCAD, SketchUp, 3ds Max, Revit, D5 Render, Blender, V-Ray, Enscape
- **Interactive and Physical Computing Tools:** Arduino, Unity 3D, KiCad, Particle, Cura, Laser Cutting, 3D Printing, CNC Machining, TouchDesigner
- **Software and AI Frameworks:** React, Three.js, Flask, LangChain, OpenAI API, RhinoCommon
- **Research and Analysis:** ClimateStudio, Ecotect, GIS, Tableau
- **Research Methods:** Systematic Literature Review (PRISMA), Qualitative Coding, LLM-Assisted Content Analysis, Survey Design, User Studies
- **Languages:** English (fluent), Mandarin (native), Japanese (basic)

Professional Experience

Product Designer May 2024 - July 2024

DesignX Company Los Altos, CA

- Developed digital educational tools for a design camp for 90 K-12 kids using Figma and Adobe Illustrator, including 12 learning modules and 5 interactive tools
- Facilitated 90 kids in building their architecture design projects by assisting with their design process and assessing their understanding of design knowledge

Architectural Designer July 2022 - October 2022

Beijing Institute of Architectural Design (BIAD) Beijing, China

- Contributed to campus planning and facade design for Peking University's new Medical School campus, producing over 20 analysis diagrams with Adobe Creative Suite, 5 renderings with Enscape and V-Ray, and 3 facade engineering drawings with AutoCAD
- Developed a Grasshopper script to rapidly modify facade arrangements and dimensions, increasing team modeling efficiency by approximately 60%